

Def: Alternative - to - transform(C):

memory-limited

Transform Architecture: Attention + Autoregression.

↓

modify architecture $\Rightarrow$ state-space model

new models

diffusion model

① not designed properly for inference
②

State-space model: (more efficient for inference)

idea: from signal processing to model long-context sequences in a sub-quadratic time.

Sq: based on classic state space models, good at synthetic long-context tasks

examples:

Mamba: allow SSMA parameters to be input-dependent; match Tq. 18.478 scale

Jamba: interleave Transformer-Mamba layers with

BASED: use linear attention + local attention (521) 1802

MiniMax-D: use linear attention + full attention

state analysis: linear + local attention (still need some full-attention) yield serious

replace OCT KV cache with OCT state $\Rightarrow$

much more efficient for inference

Diffusion models: for image/video generation, but not for text generation

random sampler $\xrightarrow{\text{denoise}}$ $\xrightarrow{\text{denoise}}$ image

idea: generate each token in parallel, refine multiple time steps